

How to: Data Science Project Report

A guide for the SoSe 2026 data project

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Outset

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- Make a selection among the methods learned in class that are **most suitable** to the problem at hand. Don't just run all algorithms and see what sticks. Argue your choice and justify it.

Report Guidelines

The following guidelines are rubric, i.e., completing all sections outlined in this presentation **will not guarantee** you a certain grade. We are providing this to help you **structure your report** and guide you to write up your project reports.

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What this guide is about

These slides cover the *report only*. *Assignment 2* is assessed by a leaderboard, not by writing — the same pipeline can serve both, but the report does not discuss *Assignment 2*.

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High mortalities following disease outbreaks can importantly reduce cultivated populations and result in economic losses. In this study, we present a continuous diphasic model (CDM) for dynamic prediction of survival of shrimp population, when affected by white spot disease (WSD). The model allows describing a smooth transition between two phases that correspond to survival before and after die-offs caused by the disease. The CDM is statistically compared with a discontinuous diphasic model (DDM) reported by us in a previous investigation.

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- Give some background if necessary including **citing original sources**.
- Clearly state what the input and output is.
- Be very **explicit**: “The input to our algorithm is an {image, amplitude, patient age, rainfall measurements, grayscale video, etc.}. We then use a {SVM, neural network, linear regression, etc.} to output a predicted {shrimp population, stock price, cancer type, music genre, etc.}.”

Comparison of first sentences of the introduction

Option A

“Natural Language Processing (NLP) has seen the development of large-scale pre-trained language models, which are trained on vast amounts of language data on unsupervised tasks (called pretraining). The model parameters obtained through pre-training are later fine-tuned on specific tasks. Famous examples of such models include BERT (Devlin et al., 2018), and more recently GPT-3 (Brown et al., 2020).”

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Option B

“In this report, we use the GTZAN dataset [1] to explore a variety of methods to classify songs into 10 genres. To build a baseline, we first implement a couple of traditional ML techniques, such as SVM and logistic regression. To improve upon the baseline, we experimented with fully connected neural networks and convolutional NN.”

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Important resources to find papers:

- Google Scholar: <https://scholar.google.com/>
- arXiv: <https://arxiv.org/>
- Semantic Scholar: <https://www.semanticscholar.org/>

Dataset and Features [$\approx 0.5 - 1$ pages]

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- Talk about the **features** you used. If you extracted features using Fourier transforms, word2vec, histogram of oriented gradients (HOG), PCA, ICA, etc. make sure to talk about it.

Comparison dataset section



Option A

“In this project, the data set is from Kaggle, which provides a large number of labeled photos: 12,500 dogs and the same number of cats. Kaggle also provides unlabeled photos for testing in another folder. The images are of non-uniform size (averaged around 350×500) and varying image quality . . .”

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Option B

"Data was sourced from the Stanford Research Repository (STARR) data warehouse, which contains data from nearly three million de-identified patient records. Relevant clinical data captured includes demographic information, diagnoses, vital signs, lab values, orders, and procedure histories. Data is imported from the physician-facing EHR (EPIC) and transformed into tables that can be queried by researchers . . ."

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- Why did you choose this algorithm/method? Allude to **properties** that are desirable given the task/data at hand.

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Option A

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Option B

"We choose a random forest because (i) the scatterplots in Figure 1 show non-linear feature–target relations, and (ii) features differ in scale by three orders of magnitude (e.g. age vs. serum creatinine). Tree splits handle both without manual scaling."

Results [$\approx$ 1 – 3 pages]

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- Make sure to discuss the figures/tables in your main text throughout this section.

Option A

“We use simple feed-forward networks with 3 hidden layers on the MNIST dataset and a final output layer with 10 units (corresponding to the number of prediction classes). The output logits are then passed through a softmax layer before the cross-entropy loss with the labels is computed. Such a rudimentary model was employed to ensure that the experiments are more controlled and the results can be explained with fewer assumptions. [...] The default metric used is test accuracy. However, to investigate the purported regularization effects, we also consider various measures of variance ...”

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Option B

"We first begin with a search for the optimal parameters of our doc2vec model. Here, we train on the price change of the RUSSELL 3000, an index representing the 3,000 largest companies by market capitalization that are incorporated in America. The motivation behind this decision was to use a security that represents the wide range of American companies that may be sensitive to Trump tweets. Figures 1 and 2 exhibit our experimental results when varying embedding vector size and number of training epochs. ..."

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- Which algorithms were the highest performing?
- Why do you think that some algorithms worked better than others?
- For future work, if you had more time, more team members, or more computational resources, what would you explore?

Comparison results section

Option A

“Generally performance was good for all types of classifiers tried. The best results achieving approximately 89% accuracy were the ANFIS algorithm and K-nearest neighbors with $K = 10$ for the smaller dataset with more features. While a few percent improvement from existing systems scoring just under 80% may not seem drastic, this can actually be quite substantial for a disease that affects such a high portion of the global population . . .”

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Option B

“The task for this project is to predict pet adoption speed given related information such as biological details, descriptions. We use traditional machine learning techniques such as logistic regression, Naive Bayes, SVM, decision trees, random forest and gradient boosting on the categorical data, as well as deep learning algorithms on the combination of categorical and text data. Among all models, the Fully Connected model on only the categorical data achieves the highest accuracy of 0.396 on the test set. Based on the result, visual information such as photos and videos of the pet seems to have the most influence on the adoption decision. . . .”

Contributions



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- Make sure that you cite the most up-to-date version of a publication. Sometimes an arxiv version exists, but the paper was later published in a journal. Then you should cite the journal version.

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- Each table and figure needs to be **referenced** in your main text.

Example: a results table

Table 1 Test-set performance on the synthetic heart-failure cohort ($n = 360$ held-out patients). All metrics in $[0, 1]$, higher is better; **bold** marks the best model per column. AUC is the area under the ROC curve.

Model	Accuracy	Precision	Recall	F1	AUC
Logistic regression	0.769	0.735	0.669	0.700	0.838
Random forest	0.875	0.862	0.821	0.841	0.936
Gradient boosting	0.869	0.860	0.807	0.833	0.919

Reference: Table 19 through the latex commands `\label{tab:results}` and `\ref{tab:results}`.

Example: a multi-panel figure

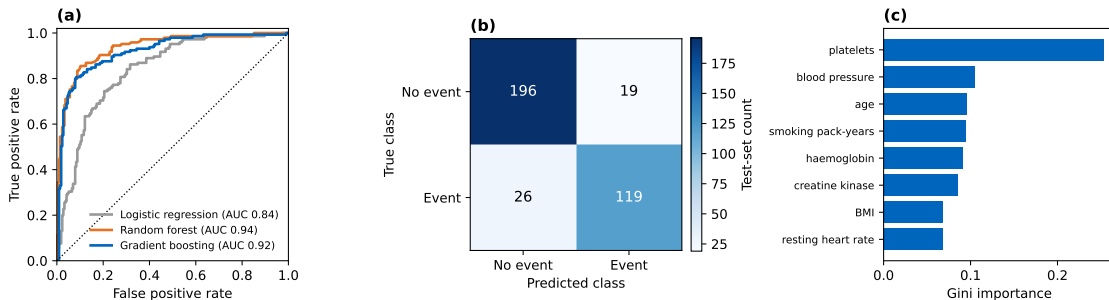


Figure 1 Evaluation of three classifiers on the held-out test set. **(a)** ROC curves; colour encodes the model and the legend reports each AUC. **(b)** Confusion matrix of the best model (random forest); cell colour and number give the test-set count. **(c)** The eight most important features of the random forest (Gini importance).

Reference: Figure 20 through the latex commands `\label{fig:results}` and `\ref{fig:results}`.

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- Use the `\appendix` command to start the appendix.

General advice



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Resources for orientation: Stanford University CSS229 projects
<https://cs229.stanford.edu/proj2021spr/>

You may use **all tools** that help you complete the assignments **including LLMs**. Be very aware though that you carry full responsibility for the content of your report. We may check individually that only content which you **fully understand** is included in your report. If we suspect and subsequently establish that you have not **fully understood** your report, we may **reject** your report.